Hexadecimal & BCD · AS · Practice

U04 · 24 written + 6 pen-first = 30 marks · Name: _____ Due: _____

All questions are authored. BCD means 8421 binary-coded decimal. Show working where requested.

1. Ours: conversion table · exam-shape rehearsal

Complete the table. One mark per complete value. **[4]**

Statement	Answer
Hexadecimal 3D represented in denary	
Denary 202 represented in hexadecimal	
Denary 84 represented in 8421BCD	
Binary 1110 0101 represented in hexadecimal	

2. Ours: a zero in the middle

Convert hex **C09** to denary. Show your working. **[2]**

3. Ours: work in the other direction

Convert denary **427** to hexadecimal. Show your working. **[2]**

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4. Ours: keep the decimal digits

(a) Represent denary **406** in BCD. **[1]**

(b) Give the denary value represented by BCD **0010 1000 0001**. **[1]**

(c) A stored BCD value is written as **1001 1011**. Identify the invalid nibble and explain why it cannot encode a decimal digit. **[2]**

5. Ours: two meanings for the bits

Represent denary **74** as (i) 8-bit ordinary binary and (ii) BCD. Explain why the encodings differ. **[3]**

6. Ours: keep all eight positions

Write hex **0A** as an 8-bit binary value. **[1]**

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7. Ours: applications

(a) Describe one practical application of BCD. State how the digit encoding is used, not just a device name. **[2]**

(b) Describe one practical application of hexadecimal. Explain its benefit in that application. **[2]**

8. Ours: a display claim

A classmate says: “A BCD nibble can directly control seven display segments, so no decoding circuitry is needed.” Correct the claim and explain what the circuitry does. **[2]**

9. Ours: compact does not mean compressed

Explain why displaying a byte as two hex digits uses fewer symbols than eight binary digits without losing information. **[2]**

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10. Ours: build a translator · pen first [6]

Design before code. Translate a binary string into hex by grouping bits and using the nibble table. No built-in base-conversion functions. These six marks are human marked on paper.

(a) Draw a flowchart or write numbered steps for binary-to-hex conversion. Include what happens when the bit count is not a multiple of four. **[2]**

(b) Trace your method for binary **110111001**. Show the padded groups and the hex output. **[2]**

(c) Translate hex **0F** into binary, preserving four bits per supplied hex digit. Explain why the first four bits are zero. **[2]**

Optional coding challenge · 0 marks. Implement both directions in **challenge_U04_HexTranslator.py**. Pen first; code when ready. The file supplies the contract and local checks. No int(), hex() or bin() in your converter functions.

Above and beyond · unmarked. In RGB colour **#2864B4**, find the three denary channel values. Increase only the blue channel by 16 (denary); write the new colour code. This is an 8-bit-per-channel RGB representation, not every colour format.

Additional space - label the question number.